

Database Technologies – Assignment 3

1. Write a query to get unique department ID from employee table.

```
KD1_Rosh_98811> USE hr
Database changed
KD1_Rosh_98811> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| hr         |
+-----+
1 row in set (0.005 sec)

KD1_Rosh_98811> SELECT DISTINCT department_id FROM employees;
+-----+
| department_id |
+-----+
| NULL          |
| 10            |
| 20            |
| 30            |
| 40            |
| 50            |
| 60            |
| 70            |
| 80            |
| 90            |
| 100           |
| 110           |
+-----+
12 rows in set (0.351 sec)
```

2. Write a query to get all the employee's details from the employees table displaying the first names in descending order.

```
KD1_Rosh_98811> SELECT * FROM employees
-> ORDER BY first_name DESC;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
180	Winston	Taylor	WTAYLOR	1.650.555.0145	2016-01-24	SH_CLERK	3200.00	NULL	120	50
171	William	Smith	WSMITH	44.1632.960026	2017-02-23	SA_REP	7400.00	0.15	148	80
206	William	Gietz	WGIEZT	1.515.555.0171	2012-06-07	AC_ACCOUNT	8300.00	NULL	205	110
195	Vance	Jones	VJONES	1.650.555.0169	2017-03-17	SH_CLERK	2800.00	NULL	123	50
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.00	NULL	103	60
141	Trenna	Rajs	TRAJS	1.650.555.0141	2013-10-17	ST_CLERK	3500.00	NULL	124	50
132	TJ	Olson	TJOLSON	1.650.555.0132	2017-04-10	ST_CLERK	2100.00	NULL	121	50
190	Timothy	Venzl	TVENZL	1.650.555.0155	2016-07-11	SH_CLERK	2900.00	NULL	122	50
170	Taylor	Fox	TFOX	44.1632.960025	2016-01-24	SA_REP	9600.00	0.20	148	80
203	Susan	Jacobs	SJACOBS	1.515.555.0168	2012-06-07	HR_REP	6500.00	NULL	101	40
173	Sundita	Kumar	SKUMAR	44.1632.960028	2018-04-21	SA_REP	6100.00	0.10	148	80
166	Sundar	Ande	SANDE	44.1632.960021	2018-03-24	SA_REP	6400.00	0.10	147	80
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
128	Steven	Markle	SMARKLE	1.650.555.0128	2018-03-08	ST_CLERK	2200.00	NULL	120	50
138	Stephen	Stiles	SSTILES	1.650.555.0138	2015-10-26	ST_CLERK	3200.00	NULL	123	50
117	Sigal	Tobias	STOBIAS	1.515.555.0117	2015-07-24	PU_CLERK	2800.00	NULL	114	30
116	Shelli	Baida	SBAIDA	1.515.555.0116	2015-12-24	PU_CLERK	2900.00	NULL	114	30
205	Shelley	Higgins	SHIGGINS	1.515.555.0170	2012-06-07	AC_MGR	12000.00	NULL	101	110
123	Shanta	Vollman	SVOLLMAN	1.650.555.0123	2015-10-10	ST_MAN	6500.00	NULL	100	50
150	Sean	Tucker	STUCKER	44.1632.960005	2015-01-30	SA_REP	10000.00	0.30	145	80
161	Sarath	Sewall	SSEWALL	44.1632.960016	2016-11-03	SA_REP	7000.00	0.25	146	80
192	Sarah	Bell	SBELL	1.650.555.0157	2014-02-04	SH_CLERK	4000.00	NULL	123	50
194	Samuel	McLeod	SMCLEOD	1.650.555.0159	2016-07-01	SH_CLERK	3200.00	NULL	123	50
137	Renske	Adawig	RLADWIG	1.650.555.0137	2013-07-14	ST_CLERK	3600.00	NULL	123	50
143	Randall	Matos	RMATOS	1.650.555.0143	2016-03-15	ST_CLERK	2600.00	NULL	124	50
191	Randall	Perkins	RPERKINS	1.650.555.0156	2017-12-19	SH_CLERK	2500.00	NULL	122	50
144	Peter	Vargas	PVARGAS	1.650.555.0144	2016-07-09	ST_CLERK	2500.00	NULL	124	50
152	Peter	Hall	PHALL	44.1632.960007	2015-08-20	SA_REP	9000.00	0.25	145	80
122	Payan	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN	7900.00	NULL	100	50
157	Patrick	Sully	PSULLY	44.1632.960012	2014-03-04	SA_REP	9500.00	0.35	146	80
202	Pat	Davis	PDAVIS	1.003.555.0167	2015-08-17	MK_REP	6000.00	NULL	201	20
155	Oliver	Tuvault	OTUVAULT	44.1632.960010	2017-11-23	SA_REP	7000.00	0.15	145	80
101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90
154	Nanette	Cambrault	NCAMBRAU	44.1632.960009	2016-12-09	SA_REP	7500.00	0.20	145	80
104	Nandita	Sarchand	NSARCHAN	1.650.555.0149	2014-01-27	SH_CLERK	4200.00	NULL	121	50
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_MGR	12008.00	NULL	101	100
130	Nozhe	Atkinson	MATHINSO	1.650.555.0130	2015-10-30	ST_CLERK	2800.00	NULL	121	50
134	Michael	Rogers	MR ROGERS	1.650.555.0134	2016-08-26	ST_CLERK	2900.00	NULL	122	50
201	Michael	Martinez	MMARTINE	1.515.555.0166	2014-02-17	MK_MAN	13000.00	NULL	100	100
120	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN	8000.00	NULL	100	50
164	Mattea	Marvins	MMARVINS	44.1632.960019	2018-01-24	SA_REP	7200.00	0.10	147	80
182	Martha	Sullivan	MSULLIVA	1.650.555.0147	2017-06-21	SH_CLERK	2500.00	NULL	120	50
113	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_ACCOUNT	6900.00	NULL	108	100
160	Louise	Doran	LDORAN	44.1632.960015	2015-12-15	SA_REP	7500.00	0.30	146	80

3. Write a query to get the employee ID, names (first_name and last_name), salary in ascending order of salary.

```
KD1_Rosh_98811> SELECT employee_id, first_name, last_name, salary
-> FROM employees
-> ORDER BY salary ASC;
```

employee_id	first_name	last_name	salary
132	TJ	Olson	2100.00
128	Steven	Markle	2200.00
136	Hazel	Philtanker	2200.00
127	James	Landry	2400.00
135	Ki	Gee	2400.00
119	Karen	Colmenares	2500.00
131	James	Marlow	2500.00
140	Joshua	Patel	2500.00
144	Peter	Vargas	2500.00
182	Martha	Sullivan	2500.00
191	Randall	Perkins	2500.00
118	Guy	Himuro	2600.00
143	Randall	Matos	2600.00
198	Donald	OConnell	2600.00
199	Douglas	Grant	2600.00
126	Irene	Mikkilineni	2700.00
139	John	Seo	2700.00
117	Sigal	Tobias	2800.00
130	Mozhe	Atkinson	2800.00
183	Girard	Geoni	2800.00
195	Vance	Jones	2800.00
116	Shelli	Baida	2900.00
134	Michael	Rogers	2900.00
190	Timothy	Venzl	2900.00
187	Anthony	Cabrio	3000.00
197	Kevin	Feeney	3000.00
115	Alexander	Khoo	3100.00
142	Curtis	Davies	3100.00
181	Jean	Fleaur	3100.00
196	Alana	Walsh	3100.00
125	Julia	Nayer	3200.00
138	Stephen	Stiles	3200.00
180	Winston	Taylor	3200.00
194	Samuel	McLeod	3200.00
129	Laura	Bissot	3300.00
133	Jason	Mallin	3300.00
186	Julia	Dellinger	3400.00
141	Trenna	Rajs	3500.00
137	Renske	Ladwig	3600.00
189	Jennifer	Dilly	3600.00
188	Kelly	Chung	3800.00
193	Britney	Everett	3900.00
192	Sarah	Bell	4000.00

4. Display first name and joining date of the employees who are either IT Programmer or Sales Man.

```
KD1_Rosh_98811> SELECT first_name, hire_date
-> FROM employees
-> WHERE job_id IN ('IT_PROG', 'SA_MAN');
```

first_name	hire_date
Alexander	2016-01-03
Bruce	2017-05-21
David	2015-06-25
Valli	2016-02-05
Diana	2017-02-07
John	2014-10-01
Karen	2015-01-05
Alberto	2015-03-10
Gerald	2017-10-15
Eleni	2018-01-29

10 rows in set (0.053 sec)

5. Display details of employees with employee ID 150 or 160.

```
KD1_Rosh_98811> SELECT * FROM employees
-> WHERE employee_id IN (150, 160);
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
150	Sean	Tucker	STUCKER	44.1632.960005	2015-01-30	SA_REP	10000.00	0.30	145	80
160	Louise	Doran	LDORAN	44.1632.960015	2015-12-15	SA_REP	7500.00	0.30	146	80

2 rows in set (0.012 sec)

6. Display first name, salary, commission pct, and hire date for employees with salary less than 10000.

```
KD1_Rosh_98811> SELECT first_name, salary, commission_pct, hire_date
-> FROM employees
-> WHERE salary < 10000;
```

first_name	salary	commission_pct	hire_date
Alexander	9000.00	NULL	2016-01-03
Bruce	6000.00	NULL	2017-05-21
David	4800.00	NULL	2015-06-25
Valli	4800.00	NULL	2016-02-05
Diana	4200.00	NULL	2017-02-07
Daniel	9000.00	NULL	2012-08-16
John	8200.00	NULL	2015-09-28
Ismael	7700.00	NULL	2015-09-30
Jose Manuel	7800.00	NULL	2016-03-07
Luis	6900.00	NULL	2017-12-07
Alexander	3100.00	NULL	2013-05-18
Shelli	2900.00	NULL	2015-12-24
Sigal	2800.00	NULL	2015-07-24
Guy	2600.00	NULL	2016-11-15
Karen	2500.00	NULL	2017-08-10
Matthew	8000.00	NULL	2014-07-18
Adam	8200.00	NULL	2015-04-10
Payam	7900.00	NULL	2013-05-01
Shanta	6500.00	NULL	2015-10-10
Kevin	5800.00	NULL	2017-11-16
Julia	3200.00	NULL	2015-07-16
Irene	2700.00	NULL	2016-09-28
James	2400.00	NULL	2017-01-14
Steven	2200.00	NULL	2018-03-08
Laura	3300.00	NULL	2015-08-20
Mozhe	2800.00	NULL	2015-10-30
James	2500.00	NULL	2015-02-16
TJ	2100.00	NULL	2017-04-10
Jason	3300.00	NULL	2014-06-14
Michael	2900.00	NULL	2016-08-26
Ki	2400.00	NULL	2017-12-12
Hazel	2200.00	NULL	2018-02-06
Renske	3600.00	NULL	2013-07-14
Stephen	3200.00	NULL	2015-10-26
John	2700.00	NULL	2016-02-12
Joshua	2500.00	NULL	2016-04-06
Trenna	3500.00	NULL	2013-10-17
Curtis	3100.00	NULL	2015-01-29

7. Display employees where the first name or last name starts with S.

```
KD1_Rosh_98811> SELECT employee_id, first_name, last_name
-> FROM employees
-> WHERE first_name LIKE 'S%'
-> OR last_name LIKE 'S%';
```

employee_id	first_name	last_name
166	Sundar	Ande
116	Shelli	Baida
192	Sarah	Bell
205	Shelley	Higgins
203	Susan	Jacobs
100	Steven	King
173	Sundita	Kumar
128	Steven	Markle
194	Samuel	McLeod
184	Nandita	Sarchand
111	Ismael	Sciarra
139	John	Seo
161	Sarath	Sewall
145	John	Singh
159	Lindsey	Smith
171	William	Smith
138	Stephen	Stiles
182	Martha	Sullivan
157	Patrick	Sully
117	Sigal	Tobias
150	Sean	Tucker
123	Shanta	Vollman

22 rows in set (0.411 sec)

8. Display details of jobs in the descending order of the title.

```
KD1_Rosh_98811> SELECT * FROM jobs
-> ORDER BY job_title DESC;
```

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
ST_MAN	Stock Manager	5500	8500
ST_CLERK	Stock Clerk	2008	5000
SH_CLERK	Shipping Clerk	2500	5500
SA_REP	Sales Representative	6000	12008
SA_MAN	Sales Manager	10000	20080
PU_MAN	Purchasing Manager	8000	15000
PU_CLERK	Purchasing Clerk	2500	5500
PR_REP	Public Relations Representative	4500	10500
AC_ACCOUNT	Public Accountant	4200	9000
IT_PROG	Programmer	4000	10000
AD PRES	President	20080	40000
MK_REP	Marketing Representative	4000	9000
MK_MAN	Marketing Manager	9000	15000
HR_REP	Human Resources Representative	4000	9000
FI_MGR	Finance Manager	8200	16000
AD_VP	Administration Vice President	15000	30000
AD_ASST	Administration Assistant	3000	6000
AC_MGR	Accounting Manager	8200	16000
FI_ACCOUNT	Accountant	4200	9000

19 rows in set (0.109 sec)

9. Display the details of the employees from department 30, earning the salary greater 10000 but do not receive the commission.

```
KD1_Rosh_98811> SELECT * FROM employees
-> WHERE department_id = 30
-> AND salary > 10000
-> AND commission_pct IS NULL;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30

1 row in set (0.014 sec)

10. Display the details of employees reporting to the managers with employee id 100 and 120.

```
KD1_Rosh_98811> SELECT * FROM employees
-> WHERE manager_id IN (100, 120);
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30
120	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN	8000.00	NULL	100	50
121	Adam	Fripp	AFRIPP	1.650.555.0121	2015-04-10	ST_MAN	8200.00	NULL	100	50
122	Payam	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN	7900.00	NULL	100	50
123	Shanta	Vollman	SVOLLMAN	1.650.555.0123	2015-10-10	ST_MAN	6500.00	NULL	100	50
124	Kevin	Mourgos	KMOURGOS	1.650.555.0124	2017-11-16	ST_MAN	5800.00	NULL	100	50
125	Julia	Nayer	JNAYER	1.650.555.0125	2015-07-16	ST_CLERK	3200.00	NULL	120	50
126	Irene	Mikkilineni	IMIKKILI	1.650.555.0126	2016-09-28	ST_CLERK	2700.00	NULL	120	50
127	James	Landry	JLANDRY	1.650.555.0127	2017-01-14	ST_CLERK	2400.00	NULL	120	50
128	Steven	Markle	SMARKLE	1.650.555.0128	2018-03-08	ST_CLERK	2200.00	NULL	120	50
145	John	Singh	JSINGH	44.1632.960000	2014-10-01	SA_MAN	14000.00	0.40	100	80
146	Karen	Partners	KPARTNER	44.1632.960001	2015-01-05	SA_MAN	13500.00	0.30	100	80
147	Alberto	Errazuriz	AERRAZUR	44.1632.960002	2015-03-10	SA_MAN	12000.00	0.30	100	80
148	Gerald	Cambraut	GCAMBRAU	44.1632.960003	2017-10-15	SA_MAN	11000.00	0.30	100	80
149	Eleni	Zlotkey	EZLOTKEY	44.1632.960004	2018-01-29	SA_MAN	10500.00	0.20	100	80
180	Winston	Taylor	WTAYLOR	1.650.555.0145	2016-01-24	SH_CLERK	3200.00	NULL	120	50
181	Jean	Fleaur	JFLEAUR	1.650.555.0146	2016-02-23	SH_CLERK	3100.00	NULL	120	50
182	Martha	Sullivan	MSULLIVA	1.650.555.0147	2017-06-21	SH_CLERK	2500.00	NULL	120	50
183	Girard	Geoni	GGEONI	1.650.555.0148	2018-02-03	SH_CLERK	2800.00	NULL	120	50
201	Michael	Martinez	MMARTINE	1.515.555.0166	2014-02-17	MK_MAN	13000.00	NULL	100	80

22 rows in set (0.008 sec)

11. Display the unique country_id from locations table.

```
KD1_Rosh_98811> SELECT DISTINCT country_id FROM locations;
```

country_id
AU
BR
CA
CH
CN
DE
GB
IN
IT
JP
MX
NL
SG
US

14 rows in set (0.063 sec)

12. Display all employees whose have the job_id as IT_PROG and FI_ACCOUNT.

```
KD1_Rosh_98811> SELECT * FROM employees
-> WHERE job_id IN ('IT_PROG', 'FI_ACCOUNT');
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	9000.00	NULL	108	100
110	John	Chen	JCHEN	1.515.555.0110	2015-09-28	FI_ACCOUNT	8200.00	NULL	108	100
111	Ismael	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_ACCOUNT	7700.00	NULL	108	100
112	Jose Manuel	Urman	JMURMAN	1.515.555.0112	2016-03-07	FI_ACCOUNT	7800.00	NULL	108	100
113	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_ACCOUNT	6900.00	NULL	108	100
103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60
104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.00	NULL	103	60
105	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_PROG	4800.00	NULL	103	60
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.00	NULL	103	60
107	Diana	Nguyen	DNGUYEN	1.590.555.0107	2017-02-07	IT_PROG	4200.00	NULL	103	60

10 rows in set (0.006 sec)

13. Display all countries in ascending order.

```
KD1_Rosh_98811> SELECT * FROM countries
-> ORDER BY country_name ASC;
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	20
AU	Australia	40
BE	Belgium	10
BR	Brazil	20
CA	Canada	20
CN	China	30
DK	Denmark	10
EG	Egypt	50
FR	France	10
DE	Germany	10
IN	India	30
IL	Israel	30
IT	Italy	10
JP	Japan	30
KW	Kuwait	30
ML	Malaysia	30
MX	Mexico	20
NL	Netherlands	10
NG	Nigeria	50
SG	Singapore	30
CH	Switzerland	10
GB	United Kingdom of Great Britain and Nort	10
US	United States of America	20
ZM	Zambia	50
ZW	Zimbabwe	50

25 rows in set (0.034 sec)